CONFOEDERATIO,
RESEARCH DIVISION (CRD).

DEMOGRAPHY AND SETTLEMENT OF THE POST-SOVIET SPACE.

A Regional Literature Review of Population and Settlement (1991AD - 2022AD).

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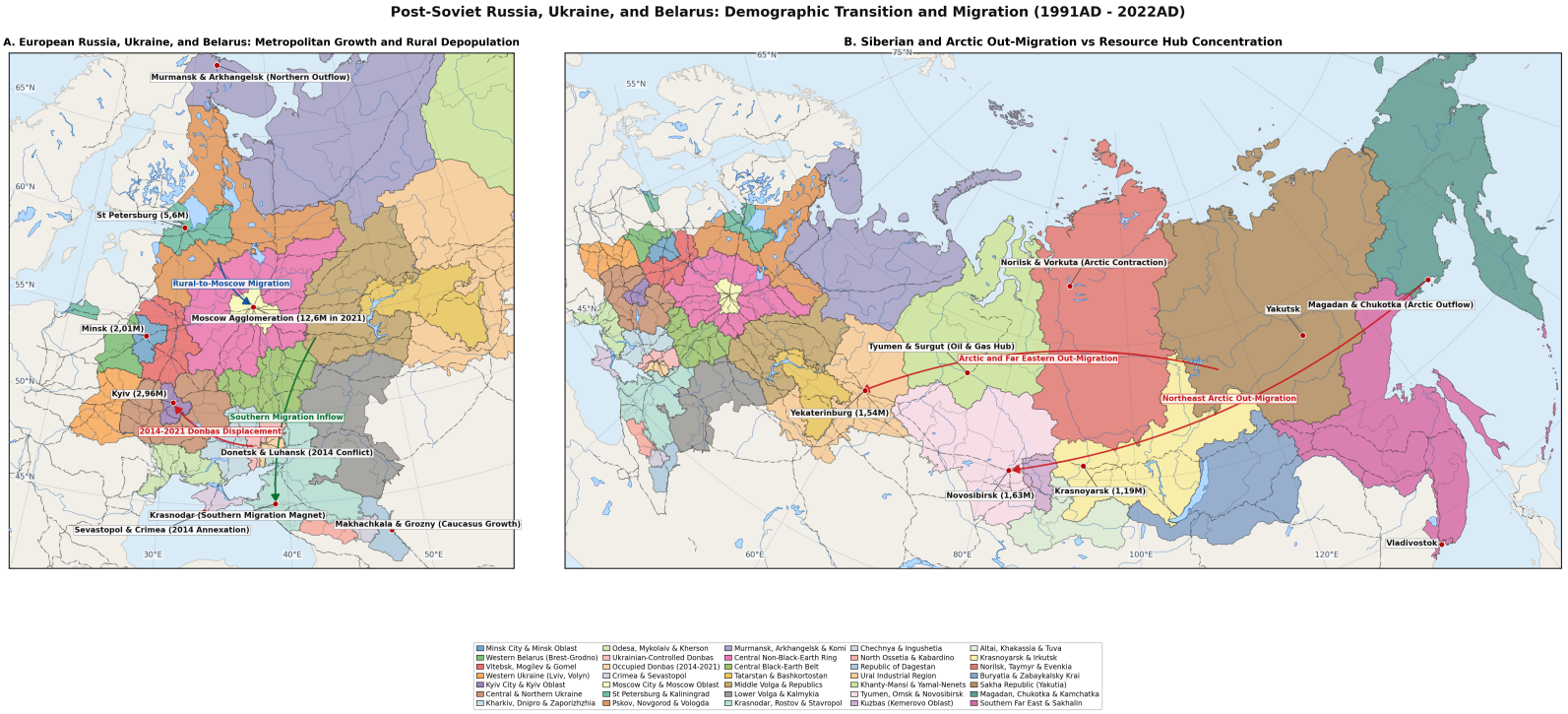


Figure 1. Thirty-five post-Soviet demographic regions, metropolitan growth magnets, northern shrinkage, armed conflicts, and migration flows (1991AD - 2021AD).

Population Estimates and Census Sources (1991-2021AD)

The dissolution of the Union of Soviet Socialist Republics in December 1991AD inaugurated a thirty-year demographic transformation across the East Slavic and Northern Eurasian space comprising the Republic of Belarus, Ukraine, and the Russian Federation (1991-2021AD). At the 1991AD transition from the late Soviet era (224 550k across the seven European and Northern Eurasian Union Republics), the restoration of independence to the Estonian (1 568k), Latvian (2 658k), Lithuanian (3 704k), and Moldavian (4 360k) Republics removed 12 290k outside the post-Soviet RU-UA-BY study envelope, leaving an exact 1991AD baseline population of 212 260k (212,26M) across the 35 demographic regions (17 969 007 km2: 10 250k across Regions 1-3 in Belarus, 51 500k across Regions 4-11 within the internationally recognised 1991AD borders of Ukraine, and 150 510k across Regions 12-35 in the Russian Federation). This total peaked briefly at 212 820k (212,82M) in 1995AD during the initial wave of post-Soviet repatriation from Central Asia and the South Caucasus [1][2][3][4]. Over the ensuing quarter-century (1995-2021AD), this territorial expanse experienced an unprecedented peacetime demographic contraction alongside an acute spatial

reordering of its urban and rural settlement hierarchy. By late 2021AD - on the eve of the full-scale Russian invasion of Ukraine in February 2022AD - the combined population across the 35 regions had fallen by -14 180k (-14,18M, or -6,7% relative to 1991AD) to 198 080k (198,08M), comprising 9 150k (-10,7%) in Belarus (Regions 1-3), 41 560k (-19,3%) across the 1991AD territories of Ukraine (Regions 4-11, including 2 210k in Government-Controlled Donbas Reg. 9, 2 470k in Pre-2022 Occupied Donbas Reg. 10, and 2 440k in Annexed Crimea and Sevastopol Reg. 11), and 147 370k (-2,1%) across the Russian Federation (Regions 12-35) [3][4][5][6].

Population series for 1991AD - 2021AD rest on three national statistical systems that diverged in census frequency and territorial coverage after 1991AD [7].

The Russian Federation (Regions 12-35, Goskomstat Rossii / Rosstat): Following the 1989AD All-Union Census (147 022k permanent residents, rising to 150 510k in the 1991AD regional baseline and 151 390k in 1995AD), Rosstat conducted three post-Soviet national enumerations: the All-Russian Population Census of 9 October 2002AD (145 167k) [8], the All-Russian Population Census of 14 October 2010AD (142 857k) [9], and the delayed All-Russian Population Census of 1 October - 14 November 2021AD (147 182k including Crimea/Sevastopol, or 147 370k within Regions 12-35) [5][10]. As demographic analysts led by Anatoly Vishnevsky, Sergei Zakharov, Evgueni Andreev, and Nikita Mkrtchyan have demonstrated, the 2002AD and 2021AD censuses relied heavily on administrative police-registration records (*registratsiya*) in metropolitan Moscow (Region 12), Saint Petersburg (Region 13), and the North Caucasus (Regions 22-24), requiring careful cross-checking against civil vital-registration ledgers (ZAGS births and deaths), intercensal retrospective adjustments, and municipal migration matrices [11][12][13][14].

Ukraine (Regions 4-11, Derzhstat Ukrainy & Ptoukha Institute): After the 1989AD Soviet census (51 452k permanent residents, rising through CIS return migration to 52,2M in 1993AD), independent Ukraine conducted only a single traditional door-to-door enumeration: the First All-Ukrainian Population Census of 5 December 2001AD (48 457k) [15]. Following the March 2014AD Russian annexation of Crimea and Sevastopol (Region 11) and the outbreak of the War in Donbas (April 2014AD), which split Donetsk and Luhansk oblasts along a 427 km Line of Contact between Government-Controlled Donbas (Region 9) and Pre-2022 Occupied Donbas (Region 10), Ukrainian demographic series were maintained through annual Ukrstat vital-registration balances [3], Ptoukha Institute reconstructions (Libanova, Gladun, Levchuk), and the December 2019AD Electronic Population Assessment (37 289k on government-controlled territory, or 41 560k across all eight Ukrainian regions Regions 4-11 in 2021AD) [6][16][17][18].

The Republic of Belarus (Regions 1-3, Minstat / Belstat): Belarus conducted full decennial enumerations on 16 February 1999AD (10 045k) [19], 14 October 2009AD (9 504k) [20], and 4-30 October 2019AD (9 413k, declining to 9 150k present population across Regions 1-3 by late 2021AD under COVID-19 excess mortality and post-2020AD emigration) [4][21][22].

The 35 Regions and Trajectories (1991-2021AD)

Rather than aggregating the post-Soviet space into broad pre-industrial ecological zones, this synthesis partitions Belarus (Regions 1-3), Ukraine (Regions 4-11), and the Russian Federation (Regions 12-35) into 35 demographic regions that explicitly separate. The three North Caucasus ethno-demographic regimes: Chechnya & Ingushetia (Region 22, First and Second Chechen Wars 1994-2009AD, Nazran IDP surge, and post-war rebound), North Ossetia (1992AD Prigorodny conflict), Kabardino-Balkaria & Karachay-Cherkessia (Region 23), and the Republic of Dagestan (Region 24, high-fertility natural increase and Makhachkala-Kaspiysk urban boom).

The three 2014-2021AD Ukrainian conflict and annexation zones: Government-Controlled Donbas (Region 9), Pre-2022 Occupied Donbas (Region 10), and Annexed Crimea & Sevastopol (Region 11).

And the ten disaggregated Asian Russian corridors and polar zones (Regions 26-35), separating the West-East Trans-Siberian and Pacific Corridors (West Siberian Trans-Siberian Belt [Reg. 27], Kuzbass Coal Basin [Reg. 28], East Siberian Trans-Siberian Corridor [Reg. 30], Buryatia & Zabaykalsky [Reg. 32], and Southern Far East & Pacific Corridor [Reg. 35]) from the Southern Siberian Indigenous Republics (Altai, Khakassia & Tuva [Reg. 29]) and the Northern Hydrocarbon/Polar Zones (Yugra & Yamal Hydrocarbon Arctic [Reg. 26], Norilsk, Taymyr & Evenkia [Reg. 31], Sakha Republic / Yakutia [Reg. 33], and Extreme Northeast Arctic Exodus: Magadan, Chukotka & Kamchatka [Reg. 34]).

Figure 2 illustrates the trajectories of ten representative regions alongside the indexed national curves (1991AD = 100), while Table 1 documents the complete benchmark series (1991AD, 1995AD, 2000AD, 2005AD, 2010AD, 2013AD, 2015AD, and 2021AD) across all 35 regions.

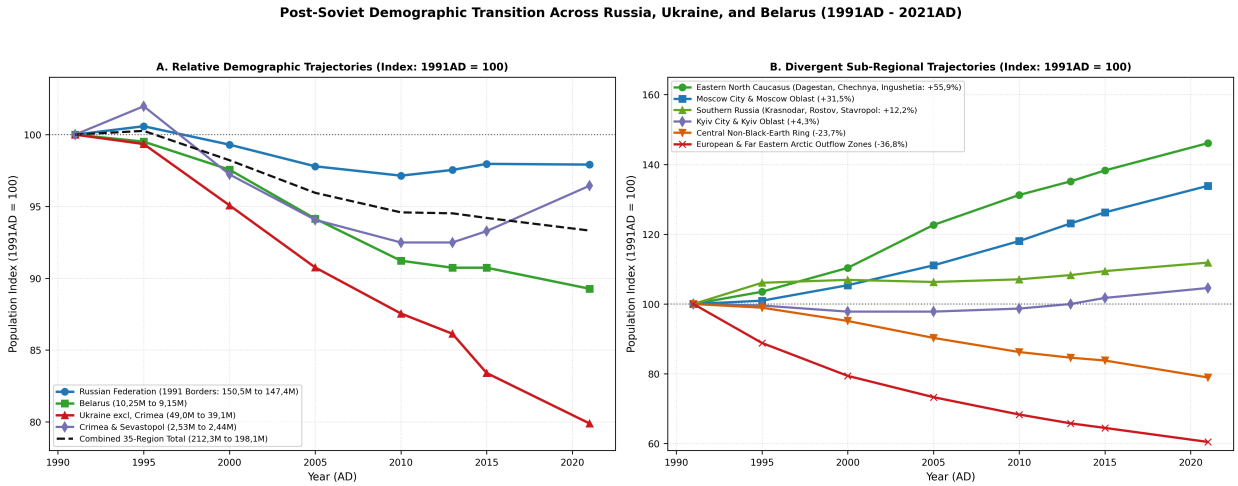


Figure 2. Regional population trajectories of expanding metropolises and contracting industrial and conflict zones alongside national indices (1991AD - 2021AD).

Post-Soviet Region (1-35)	1991AD	1995AD	2000AD	2005AD	2010AD	2013AD	2015AD	2021AD	Change
1. Minsk City & Minsk Oblast Metropolitan Magnet	3 400k	3 390k	3 360k	3 320k	3 340k	3 380k	3 420k	3 480k	+2,4%
2. Western Belarus: Brest & Grodno Oblasts	2 680k	2 690k	2 650k	2 560k	2 470k	2 440k	2 430k	2 360k	-11,9%
3. Eastern & Southeastern Belarus: Vitebsk, Mogilev & Gomel Oblasts	4 170k	4 120k	3 990k	3 770k	3 540k	3 480k	3 450k	3 310k	-20,6%
4. Western Ukraine	10 250k	10 280k	10 050k	9 780k	9 620k	9 580k	9 560k	9 320k	-9,1%
5. Kyiv City & Kyiv Metropolitan Oblast	4 580k	4 560k	4 480k	4 480k	4 520k	4 580k	4 660k	4 790k	+4,6%

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Post-Soviet Region (1-35)	1991AD	1995AD	2000AD	2005AD	2010AD	2013AD	2015AD	2021AD	Change
6. Central & Northern Ukraine Depopulation Belt	12 120k	11 990k	11 390k	10 690k	10 080k	9 770k	9 660k	9 030k	-25,5%
7. Kharkiv, Dnipropetrovsk & Zaporizhzhia Industrial East	9 100k	9 020k	8 580k	8 120k	7 780k	7 620k	7 650k	7 250k	-20,3%
8. Ukrainian Black Sea Littoral	4 700k	4 680k	4 480k	4 320k	4 220k	4 180k	4 160k	4 050k	-13,8%
9. Government-Controlled Donbas	3 620k	3 580k	3 350k	3 120k	2 950k	2 860k	2 450k	2 210k	-39,0%
10. Occupied Donbas / ORDLO	4 600k	4 540k	4 230k	3 930k	3 700k	3 590k	2 700k	2 470k	-46,3%
11. Crimea & Sevastopol	2 530k	2 580k	2 460k	2 380k	2 340k	2 340k	2 360k	2 440k	-3,6%
12. Moscow City & Moscow Oblast Post-Soviet Hyper-Centralisation Core	15 800k	15 950k	16 650k	17 550k	18 650k	19 450k	19 950k	21 150k	+33,9%
13. Saint Petersburg, Leningrad Oblast & Kaliningrad Exclave	7 620k	7 550k	7 420k	7 380k	7 520k	7 750k	7 950k	8 420k	+10,5%
14. Depopulating Northwest: Pskov, Novgorod, Vologda & Republic of Karelia	3 720k	3 660k	3 480k	3 260k	3 080k	3 010k	2 960k	2 800k	-24,7%
15. European Arctic Exodus Zone: Murmansk, Arkhangelsk, Komi & Nenets AO	4 320k	3 990k	3 630k	3 380k	3 150k	3 030k	2 970k	2 760k	-36,1%
16. Central Non-Black-Earth Mortality & Shrinkage Ring	12 350k	12 220k	11 750k	11 150k	10 650k	10 450k	10 350k	9 750k	-21,1%

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Post-Soviet Region (1-35)	1991AD	1995AD	2000AD	2005AD	2010AD	2013AD	2015AD	2021AD	Change
17. Central Black-Earth Belt	7 750k	7 980k	7 850k	7 580k	7 380k	7 320k	7 300k	7 020k	-9,4%
18. Republics of Tatarstan & Bashkortostan	7 580k	7 850k	7 890k	7 840k	7 860k	7 910k	7 950k	7 950k	+4,9%
19. Middle Volga Oblasts	15 400k	15 680k	15 350k	14 750k	14 250k	14 050k	13 950k	13 315k	-13,5%
20. Lower Volga	7 120k	7 370k	7 270k	7 040k	6 860k	6 810k	6 770k	6 440k	-9,6%
21. Southern Russian Sunbelt Magnet	13 150k	13 950k	14 060k	13 980k	14 080k	14 240k	14 390k	14 710k	+11,9%
22. Chechnya & Ingushetia	1 320k	1 180k	1 120k	1 520k	1 690k	1 790k	1 860k	2 040k	+54,5%
23. North Ossetia	1 920k	1 980k	2 080k	2 110k	2 120k	2 110k	2 110k	2 110k	+9,9%
24. Republic of Dagestan	1 880k	2 140k	2 450k	2 650k	2 910k	3 020k	3 110k	3 330k	+77,1%
25. Ural Industrial Belt	16 320k	16 470k	16 290k	15 870k	15 610k	15 540k	15 500k	15 700k	-3,8%
26. Khanty-Mansi	1 820k	1 810k	1 880k	1 990k	2 055k	2 130k	2 165k	2 255k	+23,9%
27. West Siberian Trans-Siberian Belt	11 150k	11 220k	11 030k	10 720k	10 545k	10 550k	10 595k	10 495k	-5,9%
28. Kuzbass Coal-Metallurgical Basin	3 180k	3 100k	2 980k	2 860k	2 760k	2 740k	2 720k	2 600k	-18,2%
29. Southern Siberian Republics: Altai, Khakassia & Tuva	1 090k	1 080k	1 060k	1 040k	1 045k	1 055k	1 065k	1 085k	-0,5%
30. East Siberian Trans-Siberian & Angara-Yenisei Corridor	6 030k	5 940k	5 710k	5 460k	5 315k	5 295k	5 290k	5 170k	-14,3%
31. Norilsk, Taymyr & Evenkia Polar De-Population Zone	380k	330k	290k	260k	235k	225k	220k	220k	-42,1%
32. Republic of Buryatia	2 410k	2 380k	2 290k	2 190k	2 080k	2 060k	2 050k	1 995k	-17,2%
33. Sakha Republic	1 110k	1 030k	960k	950k	958k	955k	960k	985k	-11,3%

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Post-Soviet Region (1-35)	1991AD	1995AD	2000AD	2005AD	2010AD	2013AD	2015AD	2021AD	Change
34. Extreme Northeast Arctic Exodus Zone	1 030k	770k	630k	560k	530k	515k	505k	485k	-52,9%
35. Southern Far East & Pacific Corridor	6 060k	5 760k	5 340k	5 100k	4 877k	4 805k	4 755k	4 585k	-24,3%
Sub-Total: Republic of Belarus (Regions 1-3)	10 250k	10 200k	10 000k	9 650k	9 350k	9 300k	9 300k	9 150k	-10,7%
Sub-Total: Ukraine [1991AD Borders] (Regions 4-11)	51 500k	51 230k	49 020k	46 820k	45 210k	44 520k	43 200k	41 560k	-19,3%
Sub-Total: Russian Federation [1991AD Borders] (Regions 12-35)	150 510k	151 390k	149 460k	147 190k	146 210k	146 810k	147 445k	147 370k	-2,1%
Total Post-Soviet Space (Regions 1-35)	212 260k	212 820k	208 480k	203 660k	200 770k	200 630k	199 945k	198 080k	-6,7%

Table 1. Population trajectories across the thirty-five post-Soviet regions of Belarus, Ukraine, and the Russian Federation from 1991AD to 2021AD (in thousands).

As Table 1 demonstrates, the 35-region partition exposes six structural demographic regimes across 1991AD - 2021AD [14][23][24].

Three Belarusian Demographic Zones (Regions 1-3: 10 250k -> 9 150k, -10,7%): Region 1 (Minsk City & Minsk Oblast Metropolitan Magnet) expanded by +2,4% (3 400k -> 3 480k) as Minsk city (1,61M -> 2,01M) absorbed youth migrants from across the republic; Region 2 (Western Belarus: Brest & Grodno Oblasts) contracted moderately by -11,9% (2 680k -> 2 360k); and Region 3 (Eastern & Southeastern Belarus: Vitebsk, Mogilev & Gomel Oblasts) suffered a -20,6% contraction (4 170k -> 3 310k) driven by post-Chernobyl resettlement in Gomel and Mogilev and severe rural ageing across Vitebsk [21][22].

Eight Ukrainian Demographic and Conflict Zones (Regions 4-11: 51 500k -> 41 560k, -19,3%): Only Region 5 (Kyiv City & Kyiv Metropolitan Oblast) recorded net growth (4 580k -> 4 790k, +4,6%). Region 4 (Western Ukraine: Galicia, Volhynia, Bukovina & Zakarpattia) fell by -9,1% (10 250k -> 9 320k) as higher natural fertility cushioned circular EU labour emigration; Region 8 (Ukrainian Black Sea Littoral: Odesa, Mykolaiv & Kherson) declined by -13,8% (4 700k -> 4 050k); Region 7 (Kharkiv, Dnipropetrovsk & Zaporizhzhia Industrial East) contracted by -20,3% (9 100k -> 7 250k, briefly rising from 7 620k in 2013AD to 7 650k in 2015AD as the primary reception belt for Donbas IDPs); and Region 6 (Central & Northern Ukraine Depopulation Belt) plunged by -25,5% (12 120k -> 9 030k) due to extreme rural ageing (CDR > 19-22 per 1 000 in Chernihiv and Sumy). In the conflict zones, Region 11 (Crimea & Sevastopol, annexed March 2014AD) stood at 2 440k in 2021AD (-3,6% vs 1991AD, rising from 2 340k in 2013AD via Russian military/civilian inflow), whereas the severed Donbas coal basin suffered catastrophic depopulation along

the 2014-2021AD Minsk Line of Contact: Region 9 (Government-Controlled Donbas: Mariupol, Kramatorsk, Sievierodonetsk) fell by -39,0% (3 620k in 1991AD -> 2 860k in 2013AD -> 2 210k in 2021AD) and Region 10 (Pre-2022 Occupied Donbas: Donetsk-Makiivka-Horlivka & Luhansk-Alchevsk) collapsed by -46,3% (4 600k in 1991AD -> 3 590k in 2013AD -> 2 470k in 2021AD) [16][17][25].

Metropolitan, Sunbelt, Hydrocarbon, and Petrochemical Growth Magnets in Russia (Regions 12, 13, 18, 21, and 26): Five Russian non-Caucasus regions expanded significantly: Region 12 (Moscow City & Moscow Oblast Megaregion) surged by +33,9% (15 800k -> 21 150k, +5,35M), Region 26 (Khanty-Mansi/Yugra & Yamalo-Nenets Hydrocarbon Autonomous Okrugs) by +23,9% (1 820k -> 2 255k), Region 21 (Southern Russian Sunbelt Magnet: Krasnodar, Rostov, Stavropol & Adygea) by +11,9% (13 150k -> 14 710k, +1,56M), Region 13 (Saint Petersburg, Leningrad Oblast & Kaliningrad Exclave) by +10,5% (7 620k -> 8 420k), and Region 18 (Republics of Tatarstan & Bashkortostan) by +4,9% (7 580k -> 7 950k) [24][26][27].

Disaggregated North Caucasus Republics (Regions 22, 23, and 24): Region 24 (Republic of Dagestan) recorded the highest sustained expansion in the post-Soviet space at +77,1% (1 880k -> 3 330k), driven by above-replacement fertility and the Makhachkala-Kaspiysk urban boom [28]. Region 22 (Chechnya & Ingushetia) plunged from 1 320k (1991AD) to 1 120k (2000AD) during the First and Second Chechen Wars and the permanent exodus of ethnic Russians from Grozny, before rebounding through Vainakh natural increase ($TFR \approx 2{,}5 - 2{,}8$), the Nazran IDP expansion, and federal reconstruction to 2 040k in 2021AD (+54,5%) [29][30]. Between them, Region 23 (North Ossetia-Alania, Kabardino-Balkaria & Karachay-Cherkessia) absorbed both the 1992AD Prigorodny conflict and Georgian-South Ossetian refugee inflows, growing by +9,9% (1 920k -> 2 110k).

Polar and Arctic De-Population Zones (Regions 15, 31, 33, and 34): Disaggregating the Russian North reveals the extreme gradient of the post-Soviet *Severnyy ottok* ("Northern Outflow"): Region 34 (Extreme Northeast Arctic Exodus Zone: Magadan -65%, Chukotka -70% & Kamchatka) lost more than half its population at -52,9% (1 030k -> 485k); Region 31 (Norilsk, Taymyr & Evenkia Polar De-Population Zone) fell by -42,1% (380k -> 220k); and Region 15 (European Arctic Exodus Zone: Murmansk, Arkhangelsk, Komi & Nenets AO) fell by -36,1% (4 320k -> 2 760k). By contrast, Region 33 (Sakha Republic / Yakutia) declined by only -11,3% (1 110k -> 985k, recovering from a 2005AD trough of 950k) because high indigenous Sakha fertility and the rapid growth of Yakutsk (187k -> 355k, +89,8%) partially offset the collapse of northern Russian mining towns [31][32][33].

Trans-Siberian Corridors, Southern Siberian Republics, and European Interior Shrinkage Belts (Regions 14, 16-17, 19-20, 25, 27-30, 32, and 35): In European Russia, severe ageing and rural depopulation drove Region 14 (Depopulating Northwest & Karelia) down by -24,7% (3 720k -> 2 800k) and Region 16 (Central Non-Black-Earth Mortality & Shrinkage Ring) down by -21,1% (12 350k -> 9 750k), alongside declines in Middle Volga & Finno-Ugric Republics (Region 19: -13,5%), Lower Volga & Kalmykia (Region 20: -9,6%), Central Black-Earth (Region 17: -9,4%), and the Ural Industrial Belt (Region 25: -3,8%). Across southern Siberia and the Pacific corridor, Region 29 (Southern Siberian Republics: Altai, Khakassia & Tuva) held virtually stable at -0,5% (1 090k -> 1 085k) as high indigenous Tuvan and Altai fertility ($TFR \approx 2{,}5 - 3{,}0$) offset ethnic Russian out-migration, while Region 27 (West Siberian Trans-Siberian Belt) fell by only -5,9% (11 150k -> 10 495k) thanks to the boom of Tyumen (+77,6%) and Novosibirsk (+13,7%). By contrast, coal-mine closures drove Region 28 (Kuzbass / Kemerovo Oblast) down by -18,2% (3 180k -> 2 600k), while Zapadnyy dreyf out-migration reduced Region 30 (East Siberian Trans-Siberian Corridor: Krasnoyarsk & Irkutsk) by -14,3% (6 030k -> 5 170k), Region 32 (Buryatia & Zabaykalsky Krai) by -17,2% (2 410k -> 1 995k), and Region 35 (Southern Far East & Pacific Corridor: Primorye, Khabarovsk, Amur, Jewish AO & Sakhalin) by -24,3% (6 060k -> 4 585k) [34][35][36].

The 1990s Mortality Crisis and Migration (1991-2006AD)

The defining demographic shock of the immediate post-Soviet transition (1991-2006AD) was the simultaneous collapse of period fertility and surge in adult working-age mortality across Russia, Ukraine, and Belarus - a scissors-like divergence of vital rates termed the “Russian Cross” (*Russkiy Krest*) and “Ukrainian Cross” [11][12][37][38]. Figure 3 plots the crude birth rate (CBR) and crude death rate (CDR) of Russia and Ukraine (1990-2021AD) alongside the net percentage population change across all 35 regions, while Table 2 summarises decadal vital rates, life expectancy at birth (e_0), cumulative natural decrease, and net migration balances.

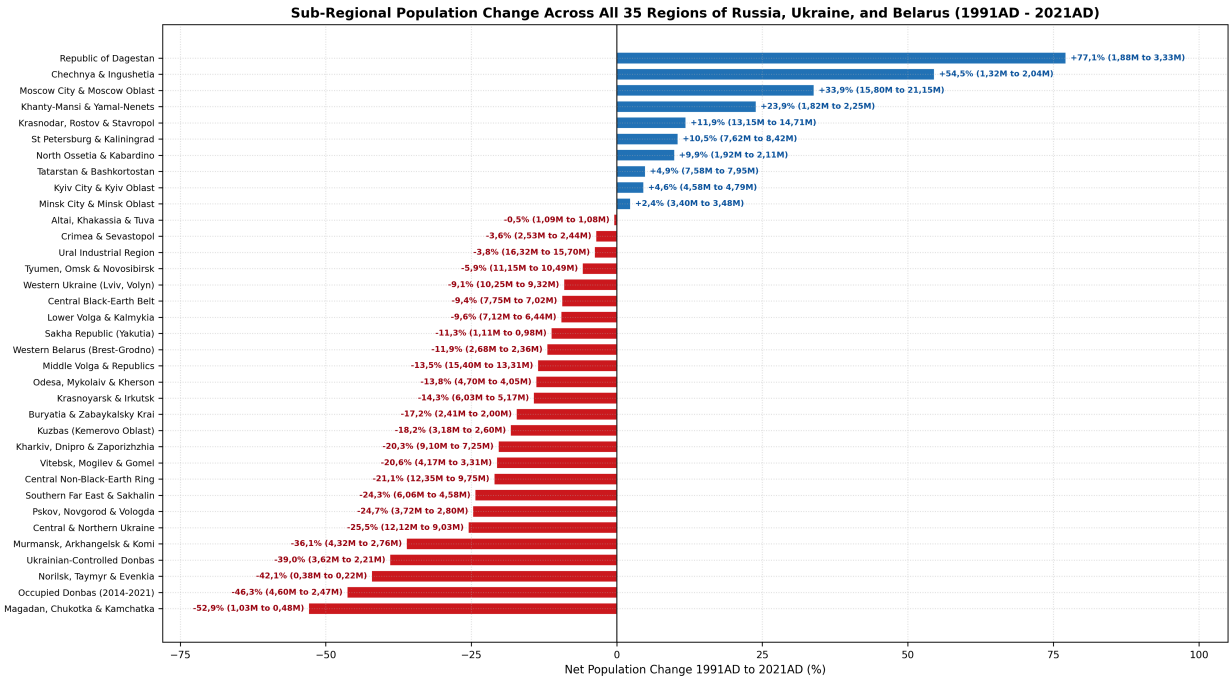


Figure 3. Crude birth and death rates per 1 000 population in Russia and Ukraine (1990AD - 2021AD) and net percentage population change across all thirty-five regions.

Country & Demographic Period	CBR / CDR (per 1 000)	Total Fertility Rate (TFR)	Male / Female e0 (Years)	Cumulative Natural Balance	Net Migration Balance
Russian Federation: Late Soviet Baseline (1989-1991AD)	13,4 / 11,2	1,89	63,8 / 74,3	+680k	+490k
Russian Federation: Acute Transition & Hyperinflation Crisis (1992-1999AD)	9,3 / 14,5	1,16 (1999AD trough)	57,6 / 71,2 (1994AD)	-5 820k	+3 650k (CIS return wave)
Russian Federation: Second Mortality Wave & Early Recovery (2000-2006AD)	9,8 / 15,8	1,30	58,9 / 72,0 (2003AD)	-5 960k	+1 180k

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Country & Demographic Period	CBR / CDR (per 1 000)	Total Fertility Rate (TFR)	Male / Female e0 (Years)	Cumulative Natural Balance	Net Migration Balance
Russian Federation: Pronatalist Recovery & Parity Window (2007-2016AD)	12,6 / 13,6	1,76 (2015AD peak)	66,5 / 77,1 (2016AD)	-1 450k (+83k in 2013-2015)	+2 650k
Russian Federation: Structural Decline & COVID-19 Shock (2017-2021AD)	10,4 / 13,9 (16,7 in 2021)	1,50	65,5 / 74,5 (2021AD)	-2 520k (-1 043k in 2021)	+980k
Ukraine [1991AD Borders]: Late Soviet Baseline (1989-1991AD)	12,6 / 12,1	1,85	65,6 / 74,9	+95k	+240k
Ukraine: Hyperinflation, Fertility Collapse & Emigration (1992-2001AD)	9,1 / 14,8	1,08 (2001AD world low)	61,2 / 72,8 (1996AD)	-2 980k	-1 150k
Ukraine: Post-2001 Depopulation & Pre-War Stabilisation (2002-2013AD)	10,1 / 15,6	1,46	62,3 / 74,1	-3 120k	-380k
Ukraine: Donbas War, Emigration & COVID-19 Shock (2014-2021AD)	9,2 / 15,4 (18,5 in 2021)	1,22 (1,16 in 2021AD)	65,2 / 74,4 (2021AD)	-2 180k (-442k in 2021)	-1 450k (Donbas/EU outflow)
Republic of Belarus: Late Soviet Baseline (1989-1991AD)	13,9 / 10,7	1,91	66,3 / 75,6	+92k	+35k
Republic of Belarus: Post-Soviet Depopulation & Rural Ageing (1992-2005AD)	9,6 / 13,8	1,25 (2003AD trough)	62,2 / 74,1 (2002AD)	-580k	+110k
Republic of Belarus: Stabilisation & COVID-19 Shock (2006-2021AD)	11,1 / 13,2 (16,4 in 2021)	1,45 (1,72 in 2015AD)	67,3 / 77,7	-480k	-95k

Table 2. Vital rates, total fertility rate, life expectancy at birth, cumulative natural balance, and net migration across Russia, Ukraine, and Belarus (1989AD - 2021AD).

Hyperinflation, Fertility Postponement, and Lowest-Low Fertility (1991AD - 2004AD)

Prior to the Soviet collapse, period fertility across all three East Slavic republics hovered near replacement ($TFR \approx 1{,}85 - 2{,}01$ in 1988-1990AD), supported by early universal marriage (mean age at first birth ~22,5 years) and state housing allocation [11][12]. Price liberalisation in January 1992AD, hyperinflation (>2 500% in Russia in 1992AD and >10 000% in Ukraine in 1993AD), the collapse of real wages by 40-60%, mass enterprise wage arrears, and the closure of municipal childcare facilities triggered an immediate collapse in second and higher-order births [18][39].

In the Russian Federation (Regions 12-35), live births fell by nearly half from 2 160k (CBR = 14,6 per 1 000)

in 1989AD to 1 215k (CBR = 8,3 per 1 000) in 1999AD, driving the total fertility rate down to an all-time low of $TFR = 1{,}157$ in 1999AD (0,92 in Saint Petersburg Region 13 and 1,01 in Moscow Region 12) [2][12].

In Ukraine (Regions 4-11), live births plummeted from 691k (13,4 per 1 000) in 1989AD to 376k (7,7 per 1 000) in 2001AD, with TFR reaching 1,08 in 2001AD - then the lowest national fertility rate recorded anywhere in the world (0,89 in Kyiv Region 5 and 0,91-0,95 across the Donbas Regions 9-10 and Industrial East Region 7) [16][18].

In Belarus (Regions 1-3), live births declined from 153k (15,0 per 1 000) in 1989AD to 88,5k (8,9 per 1 000) in 2002-2003AD, with TFR bottoming at 1,25 [20][21].

As Sergei Zakharov has shown, this fertility trough resulted from the intersection of an acute economic crisis with the onset of the Second Demographic Transition: young cohorts born in the 1970s and 1980s abandoned the Soviet pattern of universal childbearing at ages 20-23, shifting toward delayed marriage and older maternal schedules even as older cohorts halted parity progression after their first child [12][40].

The Working-Age Male Mortality Surge (1992AD - 2005AD)

Simultaneously with the collapse of births, mortality surged across all three countries between 1992AD and 1994AD, followed by a brief recovery in 1995-1998AD and a second severe mortality wave after the August 1998AD rouble default that peaked in 2003AD [13][37][41]. With the repeal of the state alcohol monopoly in 1992AD, the proliferation of cheap untaxed vodka and toxic industrial alcohol surrogates (*samogon*, colognes, and medicinal tinctures), psychosocial stress from rapid privatisation, and the deterioration of cardiovascular care produced a catastrophic surge in premature deaths among men aged 25-59 [42][43][44].

Between 1991AD and 1994AD, annual deaths in Russia rose from 1 691k (11,4 per 1 000) to 2 301k (15,7 per 1 000, exceeding 21-24 per 1 000 in Region 14 Depopulating Northwest - Pskov and Novgorod - and Region 16 Central Non-Black-Earth Shrinkage Ring - Tver, Tula, and Ivanovo), while Russian male life expectancy collapsed by -7,3 years in sixty months to 57,6 years in 1994AD (compared to 71,2 years for Russian women - a gender life-expectancy gap of 13,6 years) [13][37].

Following the 1998AD financial crash, Russian deaths climbed even higher to 2 366k (16,4 per 1 000) in 2003AD (male e0 = 58,9 years). Epidemiological case-control investigations, notably the Izhevsk Family Study (Region 19) led by David Leon, Vladimir Shkolnikov, and Evgueni Andreev, demonstrated that hazardous binge drinking of spirits and non-beverage alcohols accounted for 43% of all deaths among working-age Russian men [38][42][43].

In Ukraine, annual deaths rose from 630k (12,1 per 1 000) in 1990AD to 792k (15,2 per 1 000) in 1995-1996AD and 765k (16,2 per 1 000) in 2005-2006AD (reaching 20-23 per 1 000 across Region 6 Central & Northern Ukraine in Chernihiv and Sumy oblasts), with male life expectancy falling to 61,2 years [38][45]. Across 1992-2006AD inclusive, cumulative excess deaths over births (natural decrease) reached -11,78M in the Russian Federation, -5,62M in Ukraine, and -0,75M in Belarus - a combined vital deficit of -18,15M lives in fifteen years [2][3][20].

CIS Return Migration, Jewish Aliyah, German Aussiedler, and Ukrainian Emigration (1991-2010AD)

Why did the Russian Federation's population contract by only -2,1% across 1991-2021AD (150 510k -> 147 370k) despite over -14M in cumulative natural decrease, whereas Ukraine's population plunged by -19,3% (51 500k -> 41 560k)? The answer lies in three intersecting international migration streams [32][46][47].

The Great CIS Repatriation Wave into Russia (1992-2002AD): Between 1992AD and 2002AD, Russia recorded a net migratory gain of +4,83M permanent settlers (peaking at +845k net in 1994AD alone,

driving Russia's regional total from 150 510k in 1991AD up to 151 390k in 1995AD) from Kazakhstan (+1,85M), Central Asia (+1,42M), the South Caucasus (+820k), and the Baltics (+390k) [46][48]. These repatriates settled heavily in Region 21 (Southern Russian Sunbelt) (Krasnodar, Stavropol, Rostov: 13 150k -> 13 950k in 1995AD), Region 17 (Central Black-Earth) (Belgorod, Voronezh: 7 750k -> 7 980k in 1995AD), Region 19 (Middle Volga) (Samara: 15 400k -> 15 680k), Region 20 (Lower Volga) (Saratov, Volgograd: 7 120k -> 7 370k), and Region 27 (West Siberian Trans-Siberian Belt) (Novosibirsk, Omsk, Altai Krai: 11 150k -> 11 220k), offsetting ~55% of Russia's natural population decline during the 1990s.

Post-Soviet Jewish *Aliyah* and Ethnic German *Aussiedler* Emigration (1989-2005AD): Simultaneously, the lifting of Soviet exit restrictions triggered the permanent emigration of >1,15M post-Soviet Jews (*Aliyah* to Israel alongside emigration to the United States and Germany) and >1,85M ethnic Germans (*Spätaussiedler* to the Federal Republic of Germany) [32][46]. Across Ukraine (Regions 5-8), the Jewish population fell from 486k (1989AD) to 103k (2001AD Census), particularly in Kyiv (Reg. 5), Odesa (Reg. 8), Kharkiv (Reg. 7), and Dnipro (Reg. 7); in Belarus (Regions 1 and 3), from 112k (1989AD) to 28k (1999AD); and in Russia, from 551k (1989AD) to 230k (2002AD) and 83k (2021AD), including the emigration of over half the Jewish population of Birobidzhan / Jewish Autonomous Oblast (Region 35) [1][8][15]. Concurrently, over 450k ethnic Russian Germans emigrated directly from the German national districts of Omsk Oblast (Azovo) and Altai Krai (Halbstadt) in Region 27, Saratov and Volgograd (Region 20), and the Urals (Region 25) to Germany, while another >900k Kazakhstan Germans transited through West Siberia and the Volga [46].

Ukrainian Return Wave (1991-1993AD) Followed by Structural Emigration (1994-2021AD): Ukraine initially experienced a brief positive net migration inflow of +540k in 1991-1993AD - including ~250k Crimean Tatars returning from Uzbek deportation to Region 11 (Crimea) (2 530k -> 2 580k in 1995AD) [15][16]. From 1994AD onward, however, deeper economic contraction turned Ukraine into a major net emigration country, losing >1,4M residents to net emigration in 1994-2001AD and 2,0M-3,0M labour migrants across Region 4 (Western Ukraine EU Migration Corridor) and Region 6 (Central Ukraine) during the 2000s and 2010s [18][49][50].

Northern Exodus and Urban Polarisation (1991-2021AD)

Alongside aggregate natural decrease, the thirty years between 1991AD and 2021AD witnessed a profound spatial polarisation of settlement across the 35 regions of Russia, Ukraine, and Belarus [14][24][26][51]. Soviet central planning had dispersed tens of millions of workers into high-latitude Arctic mining towns, Siberian timber and defence settlements, and single-enterprise industrial towns (*monogoroda*) through northern wage coefficients (*severnye nadbavki*) and subsidised supply lines (*severnyy zavoz*) [31][34]. When price liberalisation in 1992AD exposed the true transport and heating costs of high-latitude and single-factory settlements, two complementary migratory processes reshaped the Eurasian map: (a) a sustained East-to-West and North-to-South internal exodus (*Zapadnyy dreyf* / "Western Drift" and *Severnyy ottok* / "Northern Outflow") out of Region 34 (Extreme Northeast Arctic Exodus Zone), Region 31 (Norilsk, Taymyr & Evenkia), Region 15 (European Arctic Exodus Zone), and Region 35 (Southern Far East & Pacific Corridor), and (b) centripetal metropolitan hyper-concentration into Region 12 (Moscow Megaregion), Region 13 (Saint Petersburg & Kaliningrad), Region 21 (Southern Russian Sunbelt), Region 26 (Yugra & Yamal Hydrocarbon Arctic), Region 1 (Minsk Magnet), and Region 5 (Kyiv Metropolis) [33][52][53]. Figure 4 contrasts expanding metropolitan and sunbelt poles against contracting Arctic, coal-mining, and conflict-affected cities (1989-2021AD), while Table 3 documents representative regional and urban shifts across the 35-region partition.

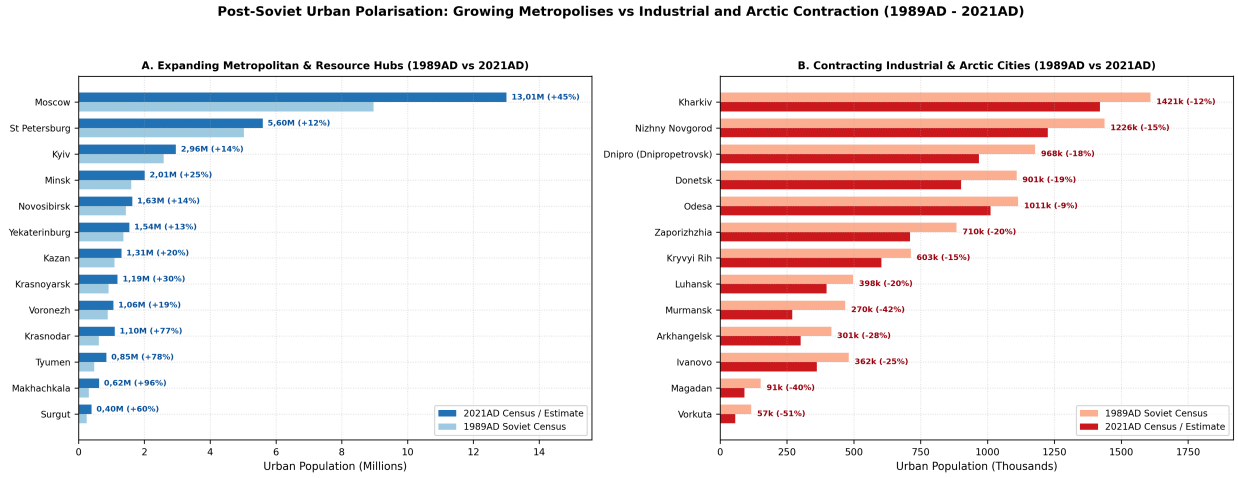


Figure 4. Divergent urban population trajectories across expanding metropolises, sunbelt capitals, and contracting Arctic and industrial cities (1989AD - 2021AD).

Trajectory Category & Settlement / Region (1-35)	1989/1991AD	2001/2002AD	2010AD	2021AD	Net Change
Extreme Northeast Exodus: Magadan, Chukotka & Kamchatka (Region 34)	1 030k (Chukotka 164k, Magadan 392k)	630k (Chukotka 54k, Magadan 183k)	530k (Chukotka 51k, Magadan 157k)	485k (Chukotka 50k, Magadan 136k)	-52,9% (Kadykchan abandoned)
Polar De-Population: Norilsk, Taymyr & Evenkia (Region 31)	380k (Norilsk aggl. 265k)	290k (Norilsk 215k)	235k (Norilsk 175k)	220k (Norilsk 175k, Igarka 3,6k)	-42,1% (Igarka -80,7%)
European Arctic Exodus: Murmansk, Arkhangelsk, Komi & Nenets AO (Region 15)	4 320k (Murmansk 468k, Vorkuta 116k)	3 630k (Murmansk 336k, Vorkuta 85k)	3 150k (Murmansk 307k, Vorkuta 71k)	2 760k (Murmansk 270k, Vorkuta 57k)	-36,1% (Vorkuta -50,9%)
Southern Far East Corridor: Primorye, Khabarovsk, Amur & Sakhalin (Region 35)	6 060k (Vladivostok 634k)	5 340k (Vladivostok 595k)	4 877k (Vladivostok 592k)	4 585k (Vladivostok 601k)	-24,3% (Amur -27,5%)
Yakutia Indigenous Urbanisation vs Mining Shrinkage (Region 33)	1 110k (Yakutsk 187k, Tiksi 11,6k)	960k (Yakutsk 211k, Tiksi 5,9k)	958k (Yakutsk 270k, Tiksi 5,1k)	985k (Yakutsk 355k, Tiksi 4,2k)	-11,3% (Yakutsk +89,8%)

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Trajectory Category & Settlement / Region (1-35)	1989/1991AD	2001/2002AD	2010AD	2021AD	Net Change
Kuzbass Coal Restructuring: Kemerovo, Novokuznetsk & Prokopyevsk (Region 28)	3 180k (Prokopyevsk 274k)	2 980k (Prokopyevsk 225k)	2 760k (Prokopyevsk 210k)	2 600k (Prokopyevsk 178k)	-18,2% (Prokopyevsk -35,0%)
Ural Industrial Belt Monogoroda: Nizhny Tagil, Zlatoust & Berezniki (Region 25)	16 320k (N. Tagil 440k, Berezniki 201k)	16 290k (N. Tagil 390k, Berezniki 173k)	15 610k (N. Tagil 362k, Berezniki 156k)	15 700k (N. Tagil 339k, Berezniki 138k)	-3,8% (Yekaterinburg +13,0%)
Central Non-Black-Earth Shrinkage Ring: Ivanovo, Tula & Tver (Region 16)	12 350k (Ivanovo 481k, Tula 540k)	11 750k (Ivanovo 432k, Tula 481k)	10 650k (Ivanovo 408k, Tula 501k)	9 750k (Ivanovo 362k, Tula 473k)	-21,1% (Ivanovo -24,7%)
Metropolitan Hyper-Concentration: Moscow City & Oblast Megaregion (Region 12)	15 800k (Moscow city 8 967k)	17 001k (Moscow city 10 383k)	18 650k (Moscow city 11 504k)	21 150k (Moscow city 13 010k)	+33,9% (Moscow +45,1%)
Moscow Ring Boomtowns: Balashikha, Khimki, Mytishchi & Krasnogorsk (Region 12)	539k combined	586k combined	765k combined	1 221k (Balashikha 521k)	+126,5% (Balashikha +285,9%)
Baltic & Northwest Core: Saint Petersburg, Leningrad & Kaliningrad (Region 13)	7 620k (StP 5 024k)	7 420k (StP 4 661k)	7 520k (StP 4 880k)	8 420k (StP 5 602k; Murino 89k)	+10,5% (StP +11,5%)
Southern Sunbelt Magnet: Krasnodar, Sochi, Rostov & Stavropol (Region 21)	13 150k (Krasnodar 621k)	14 060k (Krasnodar 646k)	14 080k (Krasnodar 745k)	14 710k (Krasnodar 1 099k)	+11,9% (Krasnodar +77,0%)
Arctic Hydrocarbon Magnet: Surgut, Nizhnevartovsk & Novy Urengoy (Region 26)	1 820k (Surgut 248k)	1 880k (Surgut 285k)	2 055k (Surgut 307k)	2 255k (Surgut 396k, Kh.-Mansiysk 107k)	+23,9% (Surgut +59,7%)

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Trajectory Category & Settlement / Region (1-35)	1989/1991AD	2001/2002AD	2010AD	2021AD	Net Change
West Siberian Trans-Siberian Boom Poles: Tyumen & Novosibirsk (Region 27)	11 150k (Tyumen 477k, Novosib. 1 437k)	11 030k (Tyumen 511k, Novosib. 1 426k)	10 545k (Tyumen 582k, Novosib. 1 474k)	10 495k (Tyumen 847k, Novosib. 1 634k)	-5,9% (Tyumen +77,6%)
Caucasus & South Siberian High-TFR Capitals: Makhachkala (Reg. 24) & Kyzyl (Reg. 29)	1 880k Reg. 24 (Makhachkala 318k; Kyzyl 87k)	2 450k Reg. 24 (Makhachkala 462k; Kyzyl 104k)	2 910k Reg. 24 (Makhachkala 572k; Kyzyl 110k)	3 330k Reg. 24 (Makhachkala 623k; Kyzyl 125k)	+77,1% Reg. 24 (Makhachkala +95,9%)
East Slavic Capitals: Kyiv Metropolis (Region 5) & Minsk Magnet (Region 1)	7 980k (4 580k + 3 400k)	7 840k (4 480k + 3 360k)	7 860k (4 520k + 3 340k)	8 270k (4 790k + 3 480k)	+3,6% (Minsk city +25,1%)

Table 3. Spatial polarisation across the thirty-five post-Soviet regions comparing northern and industrial shrinkage with metropolitan and sunbelt growth (1989AD - 2021AD).

The Northern Outflow (*Severnyy Ottok*) and Siberian Corridors (Regions 15, 26-35)

Disaggregating Asian Russia and the European North into eleven regions (Region 15 and Regions 26-35) reveals why treating Siberia and the Far East as a single block obscures three radically different demographic trajectories [31][32][33][34].

Extreme Northeast and Polar De-Population Zones (Regions 34, 31 & 15): Across Region 34 (Extreme Northeast Arctic Exodus Zone: Magadan, Chukotka & Kamchatka), population collapsed by -52,9% from 1 030k (1991AD) to 485k (2021AD): Chukotka Autonomous Okrug lost -69,5% (164k -> 50k), Magadan Oblast lost -65,3% (392k -> 136k), and Kamchatka Krai fell from 472k to 292k (-38,1%; Petropavlovsk-Kamchatsky 269k -> 165k). Dozens of Kolyma and Chukotka mining towns - including Iultin, Valkumey, and the coal town of Kadykchan (10,3k in 1989AD, completely abandoned after its 1996AD heating-plant explosion) - were shuttered, while Susuman shrank from 17k to 4,3k and Magadan city from 152k to 90k (-40,8%) [31][34]. In Region 31 (Norilsk, Taymyr & Evenkia Polar De-Population Zone) (380k -> 220k, -42,1%), the Arctic nickel city of Norilsk contracted from 265k to 175k (-34,0%), Dudinka from 32k to 19,6k, timber port Igarka by -80,7% (18,8k -> 3,6k), and Tura in Evenkia from 7,5k to 4,4k. Similarly, Region 15 (European Arctic Exodus Zone: Murmansk, Arkhangelsk, Komi & Nenets AO) lost -1,56M residents (-36,1%, 4 320k -> 2 760k), led by Vorkuta (116k -> 57k, -50,9%), Inta (60k -> 20k, -66,7%), Murmansk (468k -> 270k, -42,3%), Apatity (88k -> 50k), Arkhangelsk (416k -> 301k), and Severodvinsk (249k -> 157k) [32][35].

Indigenous Urbanisation and High-Fertility Southern Siberian Republics (Regions 33, 29 & 32): In Region 33 (Sakha Republic / Yakutia) (1 110k in 1991AD -> 950k in 2005AD -> 985k in 2021AD, -11,3%), the exodus

of >250k ethnic Russians from northern gold, tin, and coal settlements (Tiksi 11,6k -> 4,2k, Neryungri 82k -> 57k, Ust-Nera) was largely counterbalanced by high rural Sakha (Yakut) fertility and rapid migration into the capital Yakutsk, which nearly doubled from 187k (1989AD) to 355k (2021AD, +89,8%) [14][33]. Even more strikingly, Region 29 (Southern Siberian Republics: Altai, Khakassia & Tuva) held virtually constant (1 090k -> 1 085k, -0,5%): in the Tuva Republic, where ethnic Russian out-migration during the early 1990s reduced the Russian share from 32,0% (1989AD) to 10,1% (2021AD), Russia's highest total fertility rate ($TFR \approx 2{,}7 - 3{,}0$) propelled Kyzyl up by +43,7% (87k -> 125k) alongside Gorno-Altaysk (47k -> 65k, +39,3%) and Abakan (156k -> 185k) [5]. In Region 32 (Buryatia & Zabaykalsky Krai) (2 410k -> 1 995k, -17,2%), indigenous Buryat rural-urban migration expanded Ulan-Ude from 353k to 438k (+24,1%) even as Zabaykalsky Krai (Chita, Krasnokamensk) contracted sharply.

West-East Trans-Siberian Corridors vs. Arctic Hydrocarbon Growth (Regions 26, 27, 28, 30 & 35): In Region 26 (Khanty-Mansi/Yugra & Yamalo-Nenets Hydrocarbon Autonomous Okrugs), oil and gas extraction drove population up by +23,9% (1 820k -> 2 255k), led by Surgut (248k -> 396k, +59,7%), Nizhnevartovsk (245k -> 283k), Novy Urengoy (95k -> 107k), Noyabrsk (100k), Khanty-Mansiysk (34k -> 107k, +214,7%), and Salekhard (32k -> 48k). Along the West Siberian Trans-Siberian Belt (Region 27: 11 150k -> 10 495k, -5,9%), rural shrinkage in Omsk and Altai Krai (Rubtsovsk, Biysk) was offset by the boom of Tyumen (477k -> 847k, +77,6%), Novosibirsk (1 437k -> 1 634k), and university capital Tomsk (506k -> 556k). By contrast, post-Soviet mine closures in Region 28 (Kuzbass Coal-Metallurgical Basin: Kemerovo Oblast) reduced regional population by -18,2% (3 180k -> 2 600k; Prokopyevsk 274k -> 178k, Kiselevsk 128k -> 83k, Novokuznetsk 600k -> 537k) [35]. Further east, Region 30 (East Siberian Trans-Siberian Corridor: Krasnoyarsk & Irkutsk) fell by -14,3% (6 030k -> 5 170k) despite Krasnoyarsk city crossing the million threshold (913k -> 1 188k, +30,1%), while Region 35 (Southern Far East & Pacific Corridor: Primorye, Khabarovsk, Amur, Jewish AO & Sakhalin) declined by -24,3% (6 060k -> 4 585k), with Komsomolsk-on-Amur shrinking from 318k to 238k (-25,2%) while Khabarovsk (601k -> 617k) and Vladivostok (634k -> 601k) absorbed regional rural youth [33][36].

Monogoroda Shrinkage and Metropolitan Hyper-Concentration (Regions 1, 5, 12-14, 16-21 & 25)

In European Russia, Ukraine, and Belarus, Natalya Zubarevich's "Four Russias" framework captures the stark divergence between shrinking industrial/rural peripheries and booming metropolitan cores [24][26][35][54].

Interior Shrinkage and 19 400+ Abandoned Villages (Regions 14, 16-17, 19-20 & 25): Across Region 14 (Depopulating Northwest & Karelia: -24,7%), Region 16 (Central Non-Black-Earth Mortality & Shrinkage Ring: -21,1%), and Region 19 (Middle Volga & Finno-Ugric/Chuvash Republics: -13,5%), industrial cities contracted steadily (Ivanovo 481k -> 362k, Tula 540k -> 473k, Rybinsk 251k -> 177k, Dzerzhinsk 285k -> 219k, Nizhny Novgorod 1 438k -> 1 226k), while rural "spatial compression" (*szhatie prostranstva*) left over 19 400 Russian rural villages completely uninhabited (*derevni bez naseleniya*) by 2021AD [5][55]. In Region 25 (Ural Industrial Belt: 16 320k -> 15 700k, -3,8%), the growth of Yekaterinburg (1 367k -> 1 544k) and Chelyabinsk (1 148k -> 1 190k) contrasted with steep declines in metallurgical and potash *monogoroda* (Nizhny Tagil -23,0%, Zlatoust -22,1%, Berezniki -31,3%, Orsk -30,8%).

Metropolitan Hyper-Concentration (Regions 12, 13, 21, 18, 1 & 5): Conversely, Region 12 (Moscow City & Moscow Oblast Megaregion) surged by +5,35M (+33,9%, 15 800k -> 21 150k), concentrating 14,4% of the Russian Federation's entire population within Moscow city (8,97M -> 13,01M) and its MKAD commuter belt (Balashikha 135k -> 521k [+285,9%], Podolsk 315k, Khimki 257k, Mytishchi 255k, Korolev 228k, Lyubertsy 224k, Krasnogorsk 188k, Odintsovo 181k, Domodedovo 152k) [27][52]. Region 13 (Saint Petersburg, Leningrad & Kaliningrad) grew by +10,5% (7 620k -> 8 420k; Saint Petersburg 5,60M, suburb

Murino 5k -> 89k, Kaliningrad 490k), Region 21 (Southern Russian Sunbelt) by +11,9% (13 150k -> 14 710k; Krasnodar 621k -> 1 099k [+77,0%], Sochi 466k, Stavropol 547k, Rostov-on-Don 1 142k), Region 18 (Tatarstan & Bashkortostan) by +4,9% (7 580k -> 7 950k; Kazan 1 309k, Ufa 1 145k), Region 1 (Minsk Magnet) by +2,4% (3 400k -> 3 480k; Minsk 1,61M -> 2,01M), and Region 5 (Kyiv Metropolis) by +4,6% (4 580k -> 4 790k; Kyiv 2,59M -> 2,96M, Brovary 109k, Irpin 62k, Bucha 37k).

Chechen Wars and North Caucasus Demography (Regions 22, 23 & 24)

In sharp contrast to the ageing East Slavic interior, the three disaggregated North Caucasus regions - Region 22 (Chechnya & Ingushetia), Region 23 (North Ossetia-Alania, Kabardino-Balkaria & Karachay-Cherkessia), and Region 24 (Republic of Dagestan) - experienced both the most destructive military conflicts on Russian territory since 1945AD and the fastest long-term demographic growth across the entire 35-region partition (+54,5% in Region 22, +9,9% in Region 23, and +77,1% in Region 24) [28][29][30]. Table 4 summarises the principal post-Soviet armed conflicts, casualties, and forced migration flows across both the North Caucasus (Regions 22-24, 1992-2009AD) and Ukraine (Regions 9-11, 2014-2021AD).

Conflict / Territorial Event & Region	Primary Cities & Settlement Impact	Estimated War Deaths	Forced Displacement & Regional Population Shift
East Prigorodny Conflict (Region 23 North Ossetia / Region 22 Ingushetia, Oct-Nov 1992AD)	Vladikavkaz suburbs (Kambileevskoye, Chermen, Kurtat), Beslan; Nazran & Sunzha IDP camps	600 killed (plus 334 killed in Sep 2004 Beslan school siege)	40k-60k ethnic Ingush displaced from Prigorodny District to Ingushetia; 35k South Ossetian refugees into Reg. 23
First Chechen War (Dec 1994AD - Aug 1996AD) & Interwar Ichkeria (Region 22)	Grozny (400k in 1989 -> <90k in Feb 1995 siege); Gudermes, Argun, Shali, Samashki	35 000 - 50 000+ military & civilian deaths	Region 22 fell 1 320k (1991) -> 1 180k (1995); permanent exodus of >220k ethnic Russians from Chechnya
Second Chechen War & Counter-Insurgency (Aug 1999AD - Apr 2009AD, Region 22)	Grozny (reduced to 50k-60k in Jan 2000; rebuilt to 211k by 2002, 272k by 2010); Urus-Martan	25 000 - 40 000+ military & civilian deaths	Region 22 bottomed at 1 120k (2000) -> 2 040k (2021, +54,5%); 350k Chechen IDPs fled to Ingushetia (Nazran 18k -> 125k)
Dagestan Demographic Transition & Urban Boom (Region 24, 1991-2021AD)	Makhachkala (318k->623k, +95,9%); Khasavyurt (70k->155k); Derbent (78k->125k); Kaspiysk (60k->121k)	1999 Botlikh/ Kadar incursion & low-level insurgency	Region 24 surged from 1 880k (1991) to 3 330k (2021, +77,1%) via high natural increase & mountain-to-piedmont migration
Annexation of Crimea & Sevastopol (Region 11, Feb-Mar 2014AD - 2021AD)	Sevastopol (344k in 2014 -> 548k in 2021 Census); Simferopol (341k); Kerch (155k); Yevpatoria (108k)	<10 direct deaths during Mar 2014 takeover	Region 11: 60k-100k Ukrainian/Tatar IDPs out vs 250k-800k Russian settler/ military/resident inflow (2 340k -> 2 440k)

(Continued from previous page)

Conflict / Territorial Event & Region	Primary Cities & Settlement Impact	Estimated War Deaths	Forced Displacement & Regional Population Shift
War in Donbas: Partition of Region 9 (GCA) vs Region 10 (Pre-2022 Occupied Donbas, 2014-2021AD)	427 km Contact Line: Reg. 10 Occupied (Donetsk 905k, Luhansk 399k, Makiivka, Horlivka) vs Reg. 9 Ukrainian-Held (Mariupol 431k, Kramatorsk 150k, Sievierodonetsk 101k)	14 400 killed (3 404+ civilians, 4 400 Ukrainian forces, 6 500 separatist/Russian forces)	Reg. 9 fell 2 860k (2013) -> 2 210k (2021); Reg. 10 fell 3 590k (2013) -> 2 470k (2021); 1,46M IDPs to Regs 5/7 + 1,0M to Russia

Table 4. Major armed conflicts, urban devastation, estimated war deaths, and forced displacement in the North Caucasus and Ukraine from 1992AD to 2021AD.

Region 22: Chechnya and Ingushetia (1 320k in 1991AD -> 1 120k in 2000AD -> 2 040k in 2021AD, +54,5%): At the 1989AD Soviet census, the unified Checheno-Ingush ASSR contained 1 270k inhabitants (1 320k in 1991AD), including 735k ethnic Chechens, 164k Ingush, and 294k ethnic Russians (23,1%), who were heavily concentrated in Grozny (399,6k) [1][29]. During the First (1994-1996AD) and Second (1999-2009AD) Chechen Wars, urban siege warfare devastated Grozny (<60k residents in early 2000AD), Gudermes, Argun, Shali, and Urus-Martan, driving Region 22 down to 1 180k (1995AD) and 1 120k (2000AD) [29][30]. Over 320k-350k+ Chechen IDPs crossed into Ingushetia (joining 40k-60k ethnic Ingush earlier expelled from Prigorodny District in 1992AD), causing Nazran to swell nearly sevenfold from 18,2k (1989AD) to 125,1k in the 2002AD Census (122k in 2021AD alongside new republic capital Magas 15k) [8][30]. Over 250k+ ethnic Russians permanently departed Chechnya and Ingushetia for Region 21 (Southern Russian Sunbelt), reducing the Russian share in Chechnya to 1,2% (2021AD). Conversely, post-2004AD federal reconstruction rebuilt Grozny (211k in 2002AD -> 272k in 2010AD -> 329k in 2021AD), while high Vainakh fertility ($TFR \approx 2{,}5 - 2{,}8$) and young age structure (median age ~24 years) propelled Region 22 to 2 040k (1 511k in Chechnya and 510k in Ingushetia in the 2021AD Census) [5].

Region 23: North Ossetia-Alania, Kabardino-Balkaria & Karachay-Cherkessia (1 920k -> 2 110k, +9,9%): In October-November 1992AD, the East Prigorodny armed conflict outside Vladikavkaz killed ~600 people and displaced 40k-60k ethnic Ingush from Prigorodny District into Ingushetia (Region 22), while North Ossetia (Vladikavkaz 300k -> 296k, Beslan 33k -> 36k, site of the tragic September 2004AD school siege) simultaneously absorbed >35k ethnic Ossetian refugees fleeing the 1991-1992AD South Ossetian war in Georgia [28][30]. Across Kabardino-Balkaria (Nalchik 235k -> 247k, Prokhladny 58k -> 60k) and Karachay-Cherkessia (Cherkessk 113k -> 113k), moderate indigenous natural increase offset gradual ethnic Russian out-migration, stabilising Region 23 at 2 110k (2005-2021AD).

Region 24: Republic of Dagestan (1 880k -> 3 330k, +77,1%): Alone among all 35 regions, Dagestan (Region 24) nearly doubled its population across 1991-2021AD (+1,45M), driven by sustained natural increase ($TFR \approx 1{,}9 - 2{,}4$ coupled with low adult mortality, CDR ~5,5-7,0 per 1 000) and massive internal resettlement of Avar, Dargin, Lezgin, Kumyk, and Lak mountain villagers down to the Caspian coastal plain [5][28]. Consequently, Makhachkala nearly doubled from 318k (1989AD) to 623k (2021AD, +95,9%, or >1,0M in its wider agglomeration), while Khasavyurt surged by +121,4% (70k -> 155k), Kaspiysk by +101,7% (60k -> 121k), and Derbent by +60,3% (78k -> 125k).

Crimea, the Donbas War, and COVID-19 (2014-2021AD)

The final septennium of the 1991-2021AD epoch was defined by the armed conflict and territorial partition of Crimea (Region 11) and the Donbas (Regions 9 and 10) in 2014-2021AD, followed by the catastrophic 2020-2021AD COVID-19 excess mortality wave [17][25][56][57].

Region 11: Annexation of Crimea and Sevastopol (2014AD - 2021AD)

Following the Russian annexation of the Autonomous Republic of Crimea and Sevastopol (Region 11) in February-March 2014AD, the peninsula's benchmark population shifted from 2 340k (2013AD) to 2 360k (2015AD) and 2 440k (2021AD benchmark; 2 482,5k in the October 2021AD Russian Census) [5][10]. Between 60k and 100k Ukrainian citizens and Crimean Tatars relocated as IDPs to mainland Ukraine (Regions 4, 5, and 8), whereas 250k-350k+ permanent Russian military personnel, federal civil servants, and mainland retirees (and up to 500k-800k cumulative gross arrivals including construction workers and unregistered temporary residents across 2014-2021AD) moved onto the peninsula, driving Sevastopol up from 344k (2014AD) to 548k (2021AD) alongside Simferopol (341k), Kerch (155k), and Yevpatoria (108k) [17][50].

Regions 9 and 10: The War in Donbas and the 427-Kilometre Contact Line (2014AD - 2021AD)

In April 2014AD, armed conflict erupted across Donetsk and Luhansk oblasts (6 450k combined residents in 2013AD), severing the integrated Donbas coal-and-steel conurbation along a fortified 427 km Line of Contact (Minsk II) into two distinct demographic zones [3][17][25].

Region 10 (Pre-2022 Occupied Donbas: 4 600k in 1991AD -> 3 590k in 2013AD -> 2 470k in 2021AD, -46,3%): Encompassing the occupied urban core of Donetsk (965k in 2013AD -> ~905k nominal / <850k de facto in 2021AD), Luhansk (426k -> ~399k), Makiivka (353k -> ~335k), Horlivka (256k -> ~235k), Alchevsk (111k -> 105k), Yenakiieve (82k -> 75k), and Kadiivka / Stakhanov (77k -> 72k). Between 2013AD and 2015AD alone, Region 10 lost -890k inhabitants (-24,8%) through mass flight, followed by acute natural decrease (CDR > 21 per 1 000 vs CBR < 5,5 per 1 000) that reduced its present population to 2 470k by 2021AD [17].

Region 9 (Government-Controlled Donbas: 3 620k in 1991AD -> 2 860k in 2013AD -> 2 210k in 2021AD, -39,0%): Encompassing Ukrainian-controlled Mariupol (459k -> 431k), provisional Donetsk oblast seat Kramatorsk (163k -> 150k), Sloviansk (117k -> 106k), provisional Luhansk oblast seat Sievierodonetsk (109k -> 101k), Lysychansk (104k -> 93k), Kostiantynivka (76k -> 68k), Bakhmut (77k -> 71k), and Pokrovsk (65k -> 60k).

Casualties and Mass Displacement (2014-2021AD): Up to 31 December 2021AD, the Donbas conflict killed ~14 400 people and displaced over 2,5M residents: 1,46M registered IDPs relocated within government-controlled Ukraine - above all into neighbouring Region 7 (Kharkiv, Dnipropetrovsk & Zaporizhzhia) (whose population briefly rose from 7 620k in 2013AD to 7 650k in 2015AD) and Region 5 (Kyiv Metropolis) (4 580k -> 4 660k in 2015AD) - while ~1,0M refugees and migrants moved into the Russian Federation (Region 21 Rostov/Krasnodar, Region 17 Voronezh/Belgorod, and Region 12 Moscow) and Belarus (Regions 1-3) [17][18][25].

The COVID-19 Pandemic Excess Mortality Shock (2020AD - 2021AD)

After a brief 2013-2015AD vital-rate parity window in Russia (+83k natural increase), structural birth decline returned after 2016AD as the small cohorts born in the 1990s entered childbearing age, followed in 2020-2021AD by the COVID-19 pandemic [40][56][57][58].

Russian Federation (Regions 12-35): Registered deaths surged from 1 798k (2019AD) to 2 139k (2020AD) and 2 442k (2021AD, 16,7 per 1 000), producing 1,07M - 1,15M cumulative excess deaths, driving annual natural decline to a post-WWII record of -1 043k (-1,04M) in 2021AD alone, and reducing Russia's total population across Regions 12-35 from 148 075k (2018AD) to 147 370k (2021AD) [57][58].

Ukraine (Regions 4-11) and Belarus (Regions 1-3): In Ukraine, ~185k COVID-19 excess deaths pushed annual natural decline to -442k in 2021AD (CDR = 18,5 per 1 000 vs CBR = 7,3 per 1 000), reducing Regions 4-11 from 42 330k (2018AD) to 41 560k (2021AD) [3][18]. In Belarus, ~70k-85k excess deaths and post-2020AD emigration reduced Regions 1-3 from 9 250k (2018AD) to 9 150k (2021AD), bringing the entire 35-region post-Soviet space to 198 080k (198,08M) on the eve of the February 2022AD invasion [4][56].

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